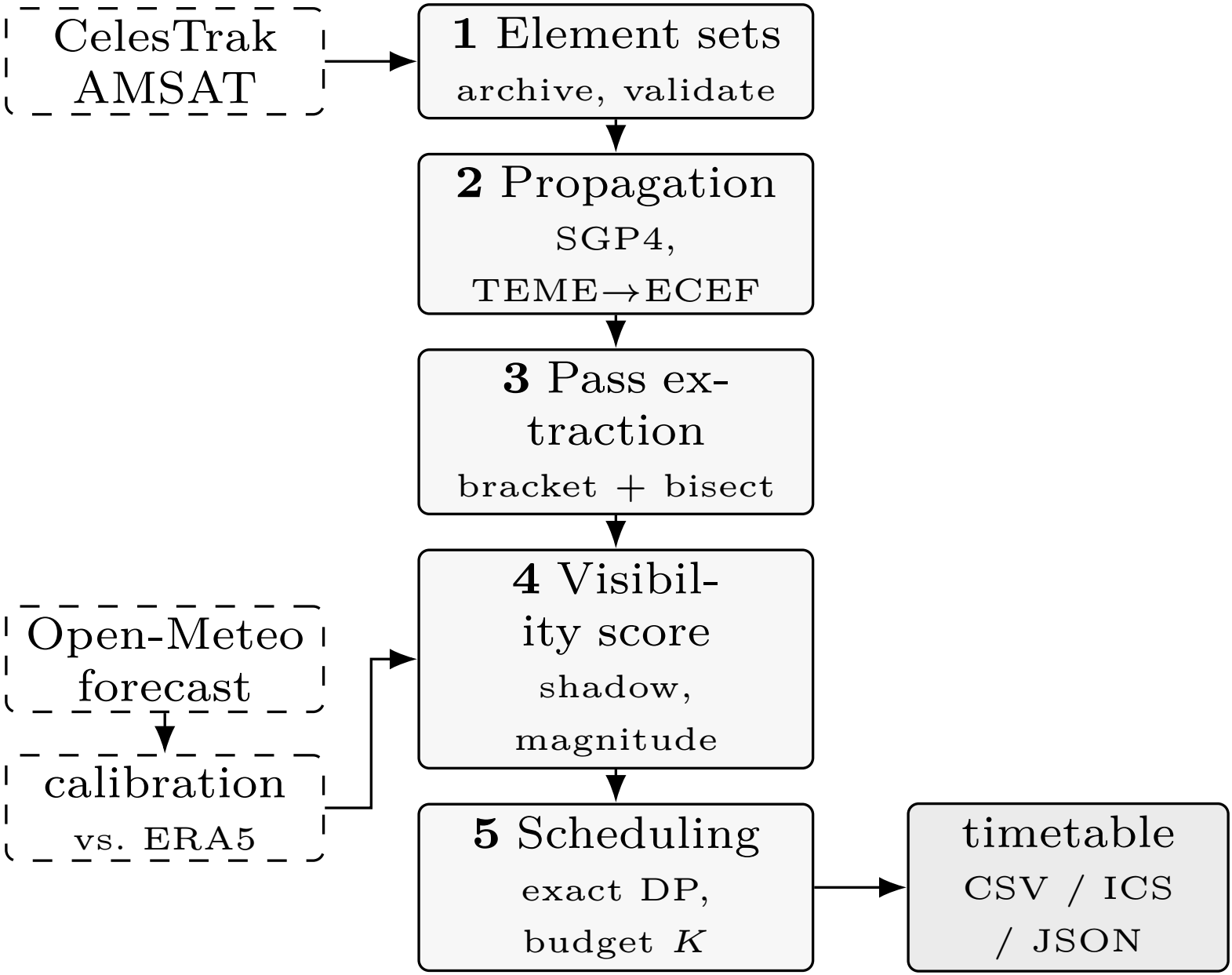
CelesTrak
AMSAT
1 Element sets
archive, validate
2 Propagation
SGP4,
TEME→ECEF
3 Pass ex-
traction
bracket + bisect
4 Visibil-
ity score
shadow,
magnitude
5 Scheduling
exact DP,
budget K
timetable
CSV / ICS
/ JSON
Open-Meteo
forecast
calibration
vs. ERA5